

SAI HARSHA VEMULA

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DevOps Engineer with expertise in AWS infrastructure automation, Terraform, and Kubernetes. Builds secure, cost-optimized CI/CD pipelines and event-driven architectures at scale. Passionate about infrastructure security, observability, and operational excellence.

Education

University of Missouri-Kansas City

Master in Computer Science

August 2023 – December 2024

Kansas City, Missouri

Vellore Institute of Technology

M Tech Integrated Software Engineering

June 2016 – June 2021

Vellore, Tamil Nadu, India

Technical Skills

Languages: Python, Shell Scripting, SQL, CSS, JavaScript

Database Technologies: MySQL, PostgreSQL, Elasticsearch, Redis

DevOps Tools & CI/CD: GitLab, Jenkins, SonarQube, Argo CD, Velero

Containerization and Orchestration: Docker, Kubernetes

AWS Cloud Services: EKS, EC2, IAM, RDS, Lambda, VPC, Cloudwatch, SNS, ECS, Step Functions, WAF

Azure Cloud Services: Azure Virtual Machines, Azure App Service, AKS, Azure Functions, Azure AD

Security & Secrets Management: HashiCorp Vault, Trivy, Gripe, GitLeaks

Protocols: TCP, UDP, IP, HTTP, HTTPS, SSH, DNS, SFTP

Monitoring & Logging: Prometheus, Grafana, Datadog, ELK Stack (Elasticsearch, Logstash, Kibana)

Infrastructure as Code (IaC): Terraform, AWS CloudFormation, Ansible

Build Tools: Maven, Gradle

Web Servers: Tomcat, Nginx, Apache HTTP Server

Operating Systems: Linux (Ubuntu, CentOS), Windows

Experience

University of Alabama – Alabama Water Institute

DevOps Engineer

July 2023 – Present

Tuscaloosa, Alabama, USA

- Architected event-driven data pipelines using **1,400+ EventBridge Schedulers** and AWS Step Functions, orchestrating National Water Model forecast executions with Terraform-managed infrastructure.
- Detected and contained a cryptocurrency mining attack within **2 hours** using AWS GuardDuty anomaly detection, preventing **\$3,000+** in unauthorized compute charges.
- Implemented organization-wide secret scanning with **Gitleaks** across **29 repositories**, automating weekly scans with GitHub Actions and Slack alerting for exposed credentials.
- Developed automated build pipelines for multi-architecture Docker images (x86 and ARM64) using GitHub Actions and AWS CodeBuild, achieving **95%** cost reduction by migrating from GitHub-hosted ARM runners.
- Implemented AWS security enhancements through CloudTrail, Security Hub, GuardDuty, and IAM Access Analyzer to ensure compliance with institutional and AWS security standards.
- Established automated vulnerability scanning using **Trivy**, **Gripe**, and **SonarQube**, integrating scans into CI/CD workflows for continuous security validation across container images.
- Implemented automated backup and retention policies for S3, EBS, and RDS resources, ensuring data resilience and compliance with university IT governance standards.
- Optimized AWS resource utilization by auditing EBS volumes, EC2 instances, and S3 storage classes, achieving **30%** monthly cost savings.
- Enhanced observability with CloudWatch metrics and dashboards, integrating SNS alerting pipelines for Step Functions workflow executions and AWS health events.
- Designed and implemented Service Control Policies (SCPs) in AWS Organizations to enforce MFA requirements, restrict resource provisioning to approved regions, and prevent accidental deletion of critical Step Functions and S3 objects.
- Integrated AWS Config and Trusted Advisor checks into continuous compliance workflows, ensuring real-time detection and remediation of policy violations across multi-account environments.
- Configured Terraform remote state management with S3 backend and DynamoDB locking, enabling secure team collaboration and preventing concurrent infrastructure modifications.
- Built automated AMI build and validation pipelines using GitHub Actions, testing across **21 VPU**s with parallel execution and automatic cleanup of failed builds.

Vitrana, Inc.

Senior Software Engineer - DevOps

July 2021 – June 2023

Bangalore, Karnataka, India

- Developed and maintained Kubernetes manifests for microservices, automating deployment and scaling workflows that improved deployment speed by **20%**.
- Applied Infrastructure as Code (IaC) using Terraform and Ansible to automate multi-environment provisioning, eliminating configuration drift and reducing setup time by **50%**.
- Established CI/CD pipelines integrating SonarQube, Trivy, and JUnit, enhancing code quality and reducing security vulnerabilities.
- Managed AWS networking and compute resources (VPC, subnets, EC2, RDS, WAF, IAM), strengthening security and reducing downtime by **20%**.
- Implemented centralized observability using Prometheus and Grafana, reducing incident response times by **40%**.
- Containerized legacy applications with Docker and improved release management using GitLab and Argo CD, streamlining deployments across environments.
- Created automated disaster recovery and backup workflows, improving data reliability and business continuity.
- Developed Python automation scripts to monitor CI/CD pipeline health and auto-trigger rebuilds on failure, reducing manual restarts and improving overall pipeline reliability by **35%**.

Vitrana, Inc.

Associate Software Engineer

August 2020 – July 2021

Bangalore, Karnataka, India

- Automated infrastructure and monitoring workflows using Bash, Python, and AWS Lambda, reducing manual intervention by **40%**.
- Led migration of on-prem applications to AWS, improving scalability and reducing operational costs by **15%**.
- Implemented IAM roles, RBAC policies, and Kubernetes access controls to enforce least privilege and ensure compliance.
- Deployed and maintained the ELK stack for centralized logging and analytics, accelerating issue resolution by **30%**.
- Developed Helm charts and automated Ansible playbooks for microservices deployment, improving consistency and release efficiency.
- Built Python utilities to automatically rotate and validate AWS IAM access keys across multiple environments, enhancing security posture and eliminating credential expiration-related outages.
- Built an analytics dashboard using AWS QuickSight and Redshift, providing real-time visibility into operational metrics.

Projects

Microservices-Based Healthcare System | Java, Spring Boot, Docker, Kubernetes, AWS

- Developed a resilient and scalable microservices-based healthcare system using Java and Spring Boot, deploying containerized services via Docker and Kubernetes for enhanced scalability and load balancing.
- Optimized CI/CD pipelines with Jenkins and GitLab, automating deployments on AWS and cutting release time by **50%**, while implementing centralized logging with the ELK stack for improved observability and issue resolution.

NextGen Research DataStream (NRDS) | AWS Step Functions, Lambda, EC2, S3, Terraform, GitHub Actions

- Architected and deployed an automated hydrological data pipeline orchestrating **1,400+ daily executions** using AWS Step Functions, Lambda, and EC2 for National Water Model forecasting workflows.
- Implemented multi-architecture CI/CD pipelines with GitHub Actions and AWS CodeBuild for ARM64/x86 Docker builds, achieving **95%** cost reduction by migrating from GitHub-hosted runners.
- Developed comprehensive health monitoring and alerting systems with CloudWatch dashboards, SNS notifications, and automated recovery workflows for production pipeline reliability.

Achievements & Certifications

AWS Certified Solutions Architect – Associate

Verified by Credly, Badge Link: [AWS Certified Solutions Architect – Associate](#)

HashiCorp Certified: Terraform Associate

Verified by Credly, Badge Link: [HashiCorp Terraform Associate](#)

Certified Kubernetes Administrator (CKA) with Practice Tests By Kodekloud

Cert Link: [Certified Kubernetes Administrator \(CKA\) with Practice Tests](#)

GitHub Actions

Cert Link: [GitHub Actions](#)

Microsoft Certified: Azure Fundamentals

Cert Link: [Microsoft Certified: Azure Fundamentals](#)